

An allosteric metabolite glue couples AICAR inversely to one-carbon availability: a theoretical prediction and its therapeutic corollaries

Alexei Vazquez^{a,*}

^a*Nodes & Links Ltd, Salisbury House, Station Road, Cambridge, CB1 2LA, UK*

ARTICLE INFO

Keywords:

one-carbon metabolism
formate
purine synthesis
AICAR
metabolite glue
PPAT
NUDT5
AMPK
thiopurines

ABSTRACT

Formate, an output of mitochondrial one-carbon metabolism, drives a switch from low to high adenine nucleotide levels in proliferating cells. The kinetic model of Oizel et al. (2020) reproduced this switch under the assumption that de novo purine synthesis is slaved to biosynthetic demand. Two independent studies have since identified the physical controller that this assumption stands in for: the ADP-ribose pyrophosphatase NUDT5 binds and inhibits phosphoribosyl pyrophosphate amidotransferase (PPAT), the rate-limiting enzyme of purine synthesis, with AMP acting as an allosteric metabolite glue and the substrate PRPP competing to dissociate the complex. We replace the demand-slaving assumption with an explicit PRPP pool and a glue-gated PPAT step, calibrated entirely against the published binding data. The model makes a sharp, falsifiable prediction: the purine precursor AICAR is coupled *inversely* to one-carbon availability, rising as one-carbon units become scarce and clearing when they are abundant. This inverse coupling is a consequence of the glue: with the feedback removed — the genetic state of a NUDT5 knockout — the model instead predicts AICAR rising with one-carbon availability, opposite in sign. The prediction reconciles the AICAR measurements of Oizel et al. (2020) without having been fitted to them, and the model independently reproduces the fall of the PRPP pool reported in NUDT5-deleted cells. We propose experiments that discriminate the glue from a constitutive throttle, and discuss the consequences for AMPK signalling and for the cytotoxicity of thiopurine and antifolate chemotherapeutics, whose action converges on this same PPAT–NUDT5 node.

1. Introduction

Proliferating cells consume one-carbon units at a high rate, principally for the de novo synthesis of purine and thymidine nucleotides. Serine is the dominant one-carbon donor: its third carbon is oxidised, chiefly in the mitochondrion, to formate, which is exported to the cytosol, condensed onto tetrahydrofolate by 10-formyl-tetrahydrofolate synthetase (FTHFS), and used to formylate the growing purine ring at two positions (Ducker and Rabinowitz, 2017; Tibbetts and Appling, 2010). When mitochondrial one-carbon output exceeds the biosynthetic demand, the excess is released as extracellular formate — “formate overflow” (Meiser et al., 2018).

Oizel et al. (2020) showed that formate is not merely a substrate but a regulator. Using a kinetic model together with metabolomics in HAP1 cells, they found that endogenous or exogenous formate induces a switch from low to high adenine nucleotide levels, raising the rate of glycolysis and repressing AMP-activated protein kinase (AMPK). Their model coupled formate to energy metabolism through purine synthesis, but treated purine synthesis itself as slaved to demand: the rate of one-carbon consumption was set equal to the biosynthetic requirement at the prevailing growth rate, with no independent enzyme kinetics. This is a perfect controller with no dynamics — adequate to reproduce the adenine nucleotide switch, but silent on the behaviour of the pathway intermediates between PRPP and the finished nucleotide.

The physical identity of that controller has now been established. Two independent studies report that the ADP-ribose pyrophosphatase NUDT5 binds phosphoribosyl pyrophosphate amidotransferase (PPAT), the committed and rate-limiting enzyme of de novo purine synthesis, and inhibits it. Witus et al. (2026) show that purine nucleotides, and AMP in particular, act as molecular glues that stabilise the inhibited PPAT–NUDT5 complex, and that the PPAT

*Corresponding author

✉ alexei.vazquez@gmail.com (A. Vazquez)
ORCID(s): 0000-0003-2764-3244 (A. Vazquez)

substrate PRPP competes at the same site to dissociate it; they term these endogenous ligands metabolite glues. Strefeler et al. (2025), screening for regulators of pyrimidine synthesis, independently identify the NUDT5–PPAT interaction, show that it preserves the PRPP pool, and report that its loss drives hyperactive purine synthesis at the expense of pyrimidines and confers resistance to nucleobase analogues. The two studies agree on the mechanism from different directions, and together they replace the demand-slaving assumption of Oizel et al. (2020) with a measured rate law.

Here we incorporate that rate law into the formate switch model. The purpose is not to improve the fit to the adenine nucleotide switch — the glue barely alters it — but to ask what the glue predicts for the pathway intermediates, where the published model was silent. The answer is a single, sharp, falsifiable statement about AICAR, and we develop its consequences for signalling and for chemotherapy.

2. Model

2.1. The published model

The working model of Oizel et al. (2020) solves for the free cytosolic concentrations of AMP, ADP, ATP and formate at steady state, from a formate balance, a one-carbon balance, an energy balance coupling ATP-consuming growth to ADP phosphorylation by glycolysis and oxidative phosphorylation, and the adenylate kinase equilibrium $[AMP][ATP] = K[ADP]^2$. The proliferation rate follows an effective Michaelis–Menten law in ATP. The step we replace is the one-carbon balance,

$$f_{CHO} = \mu \left(2[AMP] + 2[ADP] + 2[ATP] + [RNA] + \frac{5}{4}[DNA] \right), \quad (1)$$

which asserts that one-carbon consumption equals biosynthetic demand. Purine synthesis has no independent kinetics; it is defined as demand, and the free adenine pool adjusts to satisfy Eq. (1). All parameters and their literature sources are retained from Oizel et al. (2020).

2.2. A correction to the formate balance

The published implementation contains an error in the formate balance, which we correct here. The third carbon of serine reaches the folate pool by two paths: a mitochondrial path, in which serine is oxidised to free formate that FTHFS then condenses onto tetrahydrofolate, and a cytosolic path, in which SHMT1 produces CH_2 -tetrahydrofolate directly, with no free formate intermediate. The cytosolic term therefore belongs in the CHO-THF balance, but in the published code it was placed in the formate balance, making cytosolic serine catabolism an artificial source of free formate. This inflates the predicted intracellular formate and lowers the formate overflow threshold from 42% to 14% of the FTHFS rate, contrary to the conclusion of Oizel et al. (2020) that the mitochondrial path is the one required for overflow. We place the cytosolic term in the CHO-THF balance throughout. The correction does not affect the central prediction of this work (Sec. 3.2), which is evaluated with the cytosolic path inactive.

2.3. The PPAT–NUDT5 metabolite glue

Witus et al. (2026) report that the octameric PPAT₄–NUDT5₄ complex assembles through a nucleotide-independent interface; that PRPP binds the PPAT active site and dismantles the complex; and that AMP binds the same site, bridges PPAT to NUDT5, and shields the complex from PRPP-driven disassembly. AMP and PRPP therefore compete, and we write the inhibited fraction of PPAT as

$$\theta([AMP], [PRPP]) = \frac{1 + ([AMP]/K_a)^n}{1 + ([AMP]/K_a)^n + [PRPP]/K_{dP}}, \quad (2)$$

which gives a PRPP concentration for half-maximal dissociation of $IC_{50}([AMP]) = K_{dP}(1 + ([AMP]/K_a)^n)$. The genetic state of a NUDT5 knockout, in which no glue can form, is recovered by setting the numerator to zero, i.e. $\theta \equiv 0$.

We calibrate Eq. (2) entirely against the binding data of Witus et al. (2026). Their Fig. 3c reports the PRPP IC_{50} for complex dissociation at three AMP concentrations; the zero-AMP value fixes $K_{dP} = 104 \mu M$, and the shift at 0.1 and 1 mM AMP fixes $K_a = 56 \mu M$ and $n = 1.33$. As an independent check, applied to their *enzymatic* assay (Fig. 3b, a different modality) Eq. (2) predicts a half-inhibitory AMP of $332 \mu M$ against the measured $274 \pm 33 \mu M$ — a 21% error, with no parameter refitted (Fig. 1a).

2.4. Extended model

We add explicit pools for 10-formyl-tetrahydrofolate (CHO-THF), PRPP, and the two purine-pathway intermediates that flank the one-carbon-dependent formyl transfer steps: glycynamide ribonucleotide (GAR, substrate of GART) and 5-aminoimidazole-4-carboxamide ribonucleotide (AICAR, substrate of ATIC). PPAT initiates the pathway from PRPP under glue inhibition,

$$v_{\text{PPAT}} = P_{\text{max}} \frac{[\text{PRPP}]}{K_{\text{PRPP}} + [\text{PRPP}]} (1 - \theta), \quad (3)$$

and the two formyl transfers are limited by one-carbon availability $c = [\text{CHO}]/(K_c + [\text{CHO}])$,

$$v_{\text{GART}} = g_{\text{max}} \frac{[\text{GAR}]}{K_g + [\text{GAR}]} c, \quad v_{\text{ATIC}} = t_{\text{max}} \frac{[\text{AICAR}]}{K_t + [\text{AICAR}]} c. \quad (4)$$

The demand-slaving equation (1) is replaced by mass balances on the new pools and an adenine balance in which purine supply is whatever the pathway delivers, $\phi v_{\text{ATIC}} = \mu([\text{AMP}] + [\text{ADP}] + [\text{ATP}] + \Pi_{\text{ade}})$, with $\phi = 0.625$ the adenine fraction of purine flux. FTHFS is written reversibly (Appendix A), since the cytosolic synthetase domain of MTHFD1 can run backwards; this matters only for the cytosolic-serine condition and is otherwise inert. The energy balance, the adenylate kinase equilibrium and $\mu([\text{ATP}])$ are unchanged. The eight steady-state balances, all parameters with provenance tags, and the numerical method are given in Appendix A; the full parameter list is Table 2. The model has thirteen added parameters, of which three (the glue constants) are fitted to Witus et al. (2026) and the rest assumed; the sensitivity analysis of §3.5 is therefore integral, not optional.

3. Results

3.1. The glue calibration cross-validates and does not reshape the switch

Constants fitted to the dissociation assay predict the independent enzymatic K_i within 21% (Fig. 1a), supporting the competitive form of Eq. (2). Because AMP binds the same site as PRPP and stabilises the complex, rather than forming competing binary complexes, the response is monotonic in AMP with no hook effect; forward and backward parameter sweeps agree to $< 4 \times 10^{-14}$, so the model shows no hysteresis or bistability.

Across the full range of serine catabolism to formate, the inhibited fraction θ moves only from 19.5% to 21.4% (Fig. 3). The reason is quantitative: the model's free AMP spans 2.3–13.1 μM , i.e. $0.04\text{--}0.23 \times K_a$, so the AMP-sensing arm of the glue never engages and θ is pinned near its PRPP-set floor $\theta \approx 1/(1 + [\text{PRPP}]/K_{dp})$ (Fig. 1b). The glue is, in this regime, a near-constant throttle rather than a graded regulator, and the adenine nucleotide switch of Oizel et al. (2020) is essentially unchanged by it. The interest of the glue lies not in the switch but in the pathway intermediate it controls.

3.2. Key prediction: AICAR is coupled inversely to one-carbon availability

The extended model predicts that AICAR falls as one-carbon availability rises (Fig. 2a, solid). Over the scan, AICAR rises from 0.022 mM at the lowest one-carbon supply to a maximum of 0.198 mM at intermediate supply, then collapses to 0.008 mM when one-carbon units are abundant — a net inverse coupling (Spearman $\rho = -0.81$). GAR, the upstream intermediate, falls monotonically (Fig. 3).

The mechanism is transparent from the balances. At steady state the AICAR pool is set by the overshoot of PPAT initiation above the rate at which the pathway is completed,

$$[\text{AICAR}] \approx \frac{v_{\text{PPAT}} - v_{\text{ATIC}}}{\mu + k_{\text{deg}}}, \quad (5)$$

where v_{ATIC} is pinned to adenine demand through the adenine balance. When one-carbon units are scarce, ATIC — which requires a formyl-THF — stalls, PPAT continues to fire from an ample PRPP pool, and AICAR accumulates. When one-carbon units are abundant, ATIC keeps pace, the glue holds v_{PPAT} within 0.002 mM h⁻¹ of demand, and AICAR clears (Fig. 2b). The non-monotonic peak arises because at the very lowest one-carbon supply GART also stalls, so flux backs up at GAR before it can reach AICAR.

This prediction is not an artefact of fitting: no AICAR measurement entered the calibration. It nonetheless reconciles the metabolomics of Oizel et al. (2020), whose Fig. 3b reports AICAR high in one-carbon-deficient lines (MFT-SHMT1, MFT, SHMT2) and near-zero in formate-replete and wild-type lines — a significant negative association with

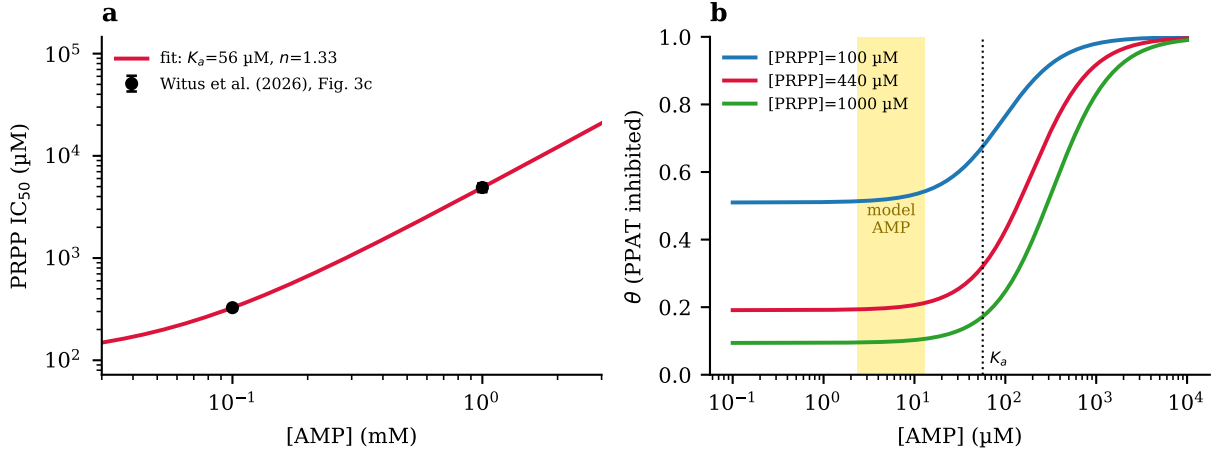


Figure 1: Calibration of the PPAT–NUDT5 glue against Witus et al. (2026). (a) PRPP IC₅₀ versus AMP; the fit to Fig. 3c predicts the independent enzymatic K_i of Fig. 3b within 21%. (b) Inhibited fraction θ versus AMP at three PRPP levels; the gold band is the free AMP range spanned by the model, lying on the flat foot of the response 4–24× below K_a .

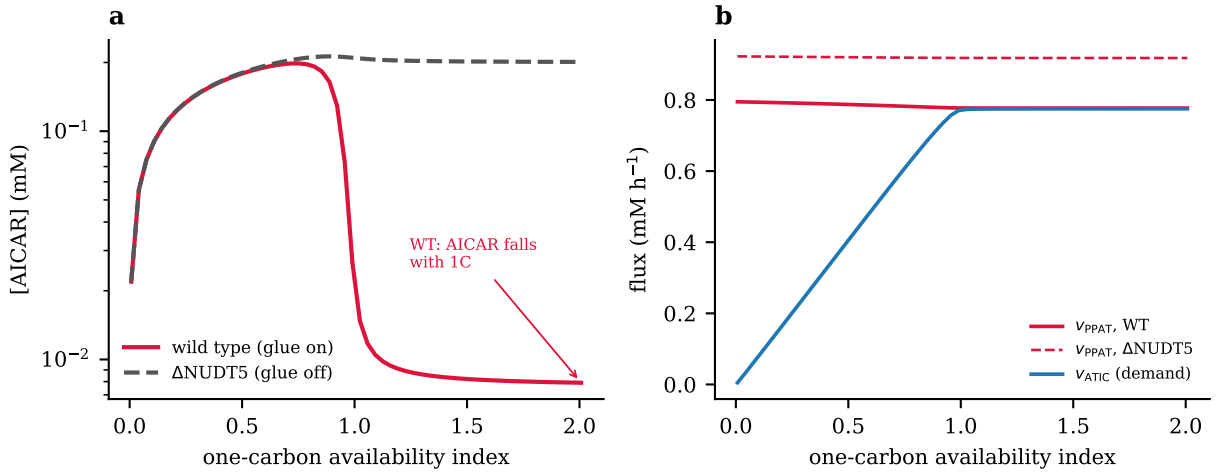


Figure 2: The key prediction and its dependence on the glue. (a) AICAR versus one-carbon availability in wild type (solid, glue on) and in a NUDT5 knockout (dashed, glue off). The two curves differ in the *sign* of the coupling. (b) PPAT initiation and ATIC completion fluxes; the glue keeps v_{PPAT} close to demand in wild type, whereas in the knockout PPAT overshoots and AICAR cannot clear.

the one-carbon availability index ($p = 5.4 \times 10^{-4}$) — and whose Fig. 5d shows, on titrating formate into MFT-SHMT1 cells, AICAR rising to a maximum near 0.25 mM formate and then falling to undetectable levels, the non-monotonic shape of Eq. (5).

3.3. The inverse coupling requires the glue: a NUDT5 knockout should reverse its sign

The inverse coupling is a property of the glue, not of the pathway topology. When the feedback is removed — the genetic state of a NUDT5 knockout, $\theta \equiv 0$ — PPAT is no longer restrained, and at high one-carbon availability it overshoots adenine demand by 0.116 mM h⁻¹ ($v_{\text{PPAT}} = 0.918$ against $v_{\text{ATIC}} = 0.802$) rather than by 0.002. By Eq. (5) AICAR then remains elevated at 0.20 mM even when one-carbon units are abundant, and the coupling turns positive (Spearman $\rho = +0.43$; Fig. 2a, dashed). The knockout is thus predicted to *invert the sign* of the AICAR–one-carbon relationship.

Table 1

A constitutive PPAT throttle mimics the glue in wild type. Scan over serine catabolism to formate; $\text{AICAR}_{\text{hi}}/\text{AICAR}_{\text{peak}}$ is the ratio of the high-one-carbon value to the peak. Only the glue additionally predicts sign inversion under NUDT5 deletion (§3.3).

P_{max} (% default)	v_{PPAT} (high)	$\text{AICAR}_{\text{hi}}/\text{AICAR}_{\text{peak}}$	collapses?
100 (glue off)	0.918	0.95	no
85	0.822	0.25	no
82	0.805	0.09	yes
80	0.787	0.05	yes
glue on (P_{max} default)	0.777	0.04	yes

Two independent observations already bracket this prediction. Strefeler et al. (2025) report that NUDT5 deletion depletes the PRPP pool, as PPAT runs unrestrained; our model reproduces this without adjustment, giving a PRPP pool 16% lower in the knockout than in wild type (Fig. 3, PRPP panel) because the same overshoot that elevates AICAR also draws down PRPP. The direction and approximate magnitude match. What has not been measured is the quantity the model most sharply predicts: the sign of the AICAR response to one-carbon supply in the knockout.

3.4. Magnitude versus mechanism, and the discriminating experiment

Because θ is near-constant, the wild-type AICAR collapse could in principle be produced by any constitutive restraint on PPAT, not specifically by the glue. We tested this by removing the glue and instead lowering the PPAT capacity P_{max} by a fixed factor (Table 1). A constitutive reduction of $\approx 18\text{--}20\%$ reproduces the AICAR collapse as well as the glue does. The wild-type data therefore constrain the *magnitude* of PPAT restraint — enough to bring v_{PPAT} within $\approx 0.03 \text{ mM h}^{-1}$ of demand — but not its *mechanism*.

Two considerations nonetheless favour the glue, and one experiment settles it. First, the required window is narrow ($\geq 18\%$) and the glue lands in it (19.5–21.4%) without being fitted to any pathway data. Second, and decisively, the two hypotheses diverge under NUDT5 deletion: removing the glue raises the overshoot to 0.116 mM h^{-1} and inverts the AICAR coupling (§3.3), whereas a cell whose PPAT capacity is constitutively 20% lower, with no glue, continues to clear AICAR normally and shows no inversion. Measuring the sign of the AICAR–one-carbon relationship in NUDT5-null cells thus discriminates the glue from a generic throttle. This is the central experimental proposal of §4.

3.5. Sensitivity to assumed parameters

Each assumed parameter was varied twofold up and down and the scan repeated. Two conclusions separate by robustness. The qualitative predictions are robust: the span of θ across the scan never exceeds 2.2 percentage points, and AICAR is elevated at low one-carbon availability in every variation, so the inverse coupling and its glue-dependence do not depend on the assumed constants. The quantitative magnitude is not robust: the absolute AICAR concentration spans a ~ 640 -fold range across the variations, since the steady-state pool is essentially $v_{\text{PPAT}}/k_{\text{deg}}$ and k_{deg} is assumed. The model predicts the *sign and shape* of the AICAR response, not its absolute level, and the proposed experiments are framed accordingly.

3.6. The glue is a dual sensor: the AMP arm reports energy charge

That θ is near-constant across the one-carbon scan reflects the regime the model occupies, not an intrinsic inertness of the feedback. The free AMP that the glue senses is not a free parameter: it is fixed by the adenylate kinase equilibrium $[\text{AMP}][\text{ATP}] = K[\text{ADP}]^2$, so it is low when the energy charge is high. In the proliferating, formate-replete state ($\text{EC} \approx 0.95$) free AMP is $2\text{--}13 \mu\text{M}$, well below $K_a = 56 \mu\text{M}$, and the AMP-sensing arm of Eq. (2) is silent; θ is set by the competing PRPP term alone. This is a property of the energy state, not of the glue.

Lowering the energy charge — by reducing the ATP-supply capacity at fixed one-carbon supply — raises free AMP steeply (Fig. 4a). It crosses K_a near $\text{EC} \approx 0.84$ and reaches $222 \mu\text{M}$ at $\text{EC} \approx 0.71$, engaging the AMP arm and driving θ from 21% to above 50% (Fig. 4b). The glue is therefore a *dual sensor*: substrate-gated through PRPP at high energy charge — the regime of the AICAR prediction above — and energy-gated through AMP once the energy charge falls. Because the AMP arm and AMPK read the same rising free AMP, the glue shuts down the ATP- and PRPP-consuming purine pathway under exactly the conditions that activate AMPK (§5.1), a coherent energy-conservation response.

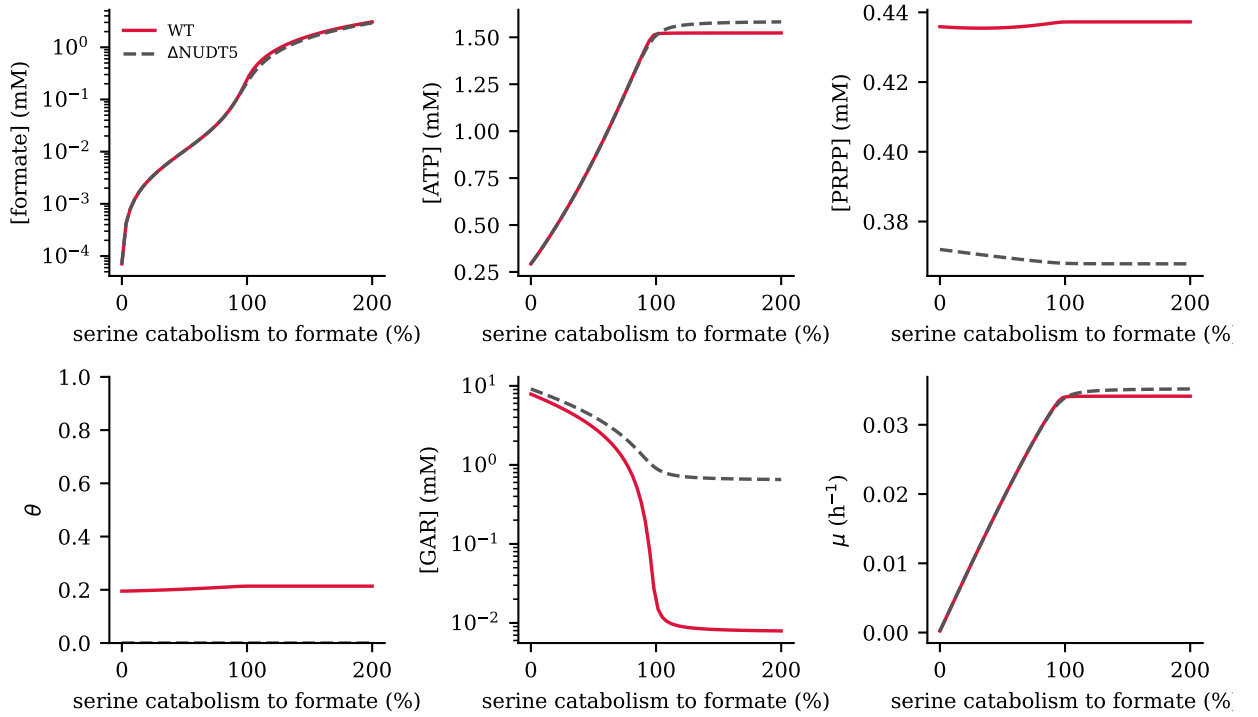


Figure 3: Extended model across serine catabolism to formate, in wild type (solid) and NUDT5 knockout (dashed). The adenine nucleotide switch (ATP) is essentially unchanged by the glue; PRPP is drawn down in the knockout, matching Strefeler et al. (2025); θ is near-constant.

This also resolves an apparent tension with measured nucleotide levels. The free AMP of the model (2–13 μ M) is far below the total AMP reported by metabolomics (10²–10³ μ M), but the two are different quantities: the glue constants were measured against free AMP, whereas metabolomics reports total, largely protein-bound AMP. A free AMP as high as 100–300 μ M would require an energy charge near 0.7 (Fig. 4a), i.e. an energy-stressed rather than a proliferating state.

4. Proposed experiments

The prediction is designed to be falsifiable with established assays. We propose, in order of decisiveness:

- 1. Sign of the AICAR–one-carbon relationship, wild type versus NUDT5-null.** Titrate formate into SHMT2-deficient cells, or a serine/glycine gradient into wild-type cells, and quantify intracellular AICAR by LC–MS across the gradient, in parallel in NUDT5-null cells. The model predicts a *negative* slope in wild type and a *positive* slope in the knockout (Fig. 2a). A sign inversion confirms the glue; an unchanged negative slope indicates a NUDT5-independent throttle and refutes the specific mechanism. This is the single experiment that separates the two hypotheses of §3.4.
- 2. Glue-pocket point mutant.** Repeat (1) in cells carrying the NUDT5 glue-interface mutation (L217A/K218A of Witus et al., 2026), which abolishes AMP-dependent inhibition while leaving NUDT5 expressed and its hydrolase activity intact. The mutant should phenocopy the knockout (positive slope), isolating the glue from any scaffolding or catalytic role of NUDT5.
- 3. PRPP and AICAR co-measurement.** The model ties the AICAR overshoot and the PRPP draw-down to the same PPAT over-firing (§3.3). Measuring both in the same samples across the one-carbon gradient tests the predicted anti-correlation between the AICAR excursion and the PRPP pool, linking our prediction to the PRPP phenotype of Strefeler et al. (2025).

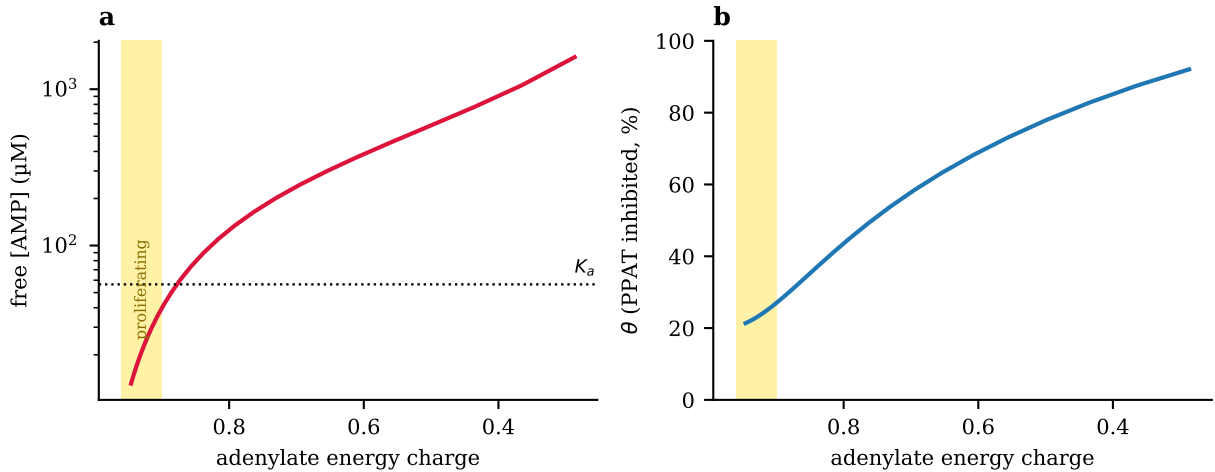


Figure 4: The AMP arm of the glue is an energy-charge sensor. Free AMP is fixed by the adenylate kinase equilibrium and rises steeply as the energy charge falls (a), crossing the glue affinity K_a and driving the inhibited fraction θ from its PRPP-set floor towards saturation (b). The gold band marks the proliferating, high-energy-charge regime in which the AICAR prediction is evaluated and the AMP arm is silent.

4. **AMPK readout.** Since AICAR is an AMP mimetic and AMPK activator (§5.1), assay phospho-AMPK (Thr172) and phospho-ACC across the gradient. The model predicts AMPK activity tracks AICAR, i.e. falls with rising one-carbon supply in wild type and fails to fall in the knockout, providing an orthogonal, antibody-based confirmation independent of metabolite quantification.

The absolute AICAR concentrations are not predicted (§3.5); all four experiments are framed as tests of *sign*, *ordering* and *monotonicity*, which are robust model outputs.

5. Discussion

5.1. AICAR as a node linking one-carbon supply to energy signalling

The prediction acquires its significance from what AICAR is. Beyond its role as a purine-pathway intermediate, AICAR (as its monophosphate, ZMP) is an AMP mimetic that allosterically activates AMPK, the master sensor of cellular energy charge (Corton et al., 1995; Hardie et al., 2012); the pharmacological AMPK activator AICA-riboside acts through precisely this species. An inverse coupling of AICAR to one-carbon availability therefore places one-carbon metabolism upstream of AMPK through a route independent of the adenylate energy charge itself.

This reframes an observation of Oizel et al. (2020), who reported that formate represses AMPK and attributed it in part to the fall of AICAR. The present model supplies the mechanism and, crucially, its controller: the fall of AICAR with rising one-carbon supply is enforced by the PPAT–NUDT5 glue, and would not occur — indeed would reverse — in its absence. The cell thus reads its one-carbon and purine-synthetic status, via the AICAR level set by the glue, into the activity of a kinase that governs glycolysis, fatty-acid oxidation and biosynthetic commitment. A NUDT5-null cell, on this account, is predicted to mis-report its one-carbon status to AMPK, sustaining an AICAR/ZMP tone — and hence an AMPK tone — that is inappropriately high when one-carbon units are plentiful. This connects a metabolite-glue, a folate-pathway input and an energy-signalling output in a single testable circuit, and may bear on the metabolic phenotypes of cells with altered NUDT5 or one-carbon-enzyme expression, including many tumours.

5.2. Corollaries for thiopurine and antifolate chemotherapy

The PPAT–NUDT5 node is not only a physiological sensor but a pharmacological target, and the model yields several corollaries for drugs that act on it.

Thiopurines. Witus et al. (2026) show that thiopurine metabolites (6-methyl-thio-IMP and related species) are themselves glues of PPAT–NUDT5, occupying the same pocket as AMP but with distinct geometry and enhanced

potency. In the model such a compound raises θ towards saturation independently of the endogenous AMP and PRPP levels. Two consequences follow. First, a thiopurine-driven increase in θ shuts PPAT and, by Eq. (5), should *lower* the AICAR overshoot rather than raise it — distinguishing the glue arm of thiopurine action from the classical antimetabolite arm (incorporation of thioguanine nucleotides into DNA), and predicting that AICAR is not a marker of this component of the response. Second, because the glue and PRPP compete, the efficacy of a thiopurine glue should depend on the cellular PRPP level, and hence on one-carbon and folate status: conditions that raise PRPP (one-carbon limitation) should antagonise glue-mediated PPAT inhibition, while conditions that lower PRPP should potentiate it.

Antifolates. Methotrexate and related antifolates deplete formyl-tetrahydrofolate. In the model this is a move to low one-carbon availability: ATIC and GART stall, PPAT continues from an ample PRPP pool, and AICAR accumulates — the well-documented “AICAR transformylase” block. The model adds that this same low-one-carbon state maximally dissociates the PPAT–NUDT5 glue (high PRPP, low AMP), consistent with the report of Witus et al. (2026) that methotrexate dissociates the endogenous complex. Antifolates and thiopurine glues thus push the node in *opposite* directions — one dissociating, one stabilising the inhibitory complex — which predicts antagonism if combined, and identifies PRPP and AICAR as candidate pharmacodynamic markers of where a given cell sits on this axis.

Resistance. Strefeler et al. (2025) report that NUDT5 loss confers resistance to nucleobase analogues that require PRPP-dependent activation (6-thioguanine, 5-fluorouracil) but not to their nucleoside forms, because unrestrained PPAT depletes the PRPP needed to activate the drug. The model recovers the mechanistic basis — the PRPP draw-down of §3.3 — and predicts a corollary: the same NUDT5-null state, by inverting the AICAR coupling, should also blunt any therapeutic strategy that relies on an AICAR/AMPK response to one-carbon restriction. Resistance to nucleobase analogues and altered AICAR/AMPK signalling are, on this view, two faces of the same lesion, and NUDT5 status is predicted to stratify both.

5.3. Limitations

The model is minimal by design. Its free AMP is an output of the adenylate kinase equilibrium and is structurally low (2–13 μM), which places the AICAR prediction in the PRPP-gated, energy-charge-silent regime (§3.6). This is a scoping condition rather than an unconstrained assumption: free AMP is fixed by the equilibrium and the high energy charge of proliferating cells, and the AMP arm engages only once the energy charge falls, so the prediction is stated for the high-energy-charge state and would need re-evaluation under energy stress. The relevant test is the adenylate energy charge, obtainable from existing nucleotide data, bearing in mind that metabolomics reports total rather than free AMP. Nine of the thirteen added parameters are assumed, and the absolute AICAR level is consequently unconstrained (§3.5); only its sign and shape are predicted. The glue also accepts IMP, GMP and AICAR itself as ligands (Witus et al., 2026), which we do not model — an AICAR-sensitive glue would be self-reinforcing at low one-carbon supply and could sharpen the predicted excursion. Finally, PPAT–NUDT5 controls purine synthesis; neither source reports a route from the complex to growth signalling, so the proliferation law is retained unchanged from Oizel et al. (2020).

6. Conclusion

Incorporating the measured PPAT–NUDT5 metabolite glue into the formate switch model, with all glue constants fixed by published binding data, yields a single sharp prediction: AICAR is coupled inversely to one-carbon availability, and this inverse coupling is enforced by the glue and should reverse its sign in a NUDT5 knockout. The prediction reconciles existing but previously unexplained AICAR measurements, is independently consistent with the PRPP phenotype of NUDT5-null cells reported by a second laboratory, and is falsifiable by a direct experiment that distinguishes the glue from a generic throttle. Its corollaries reach into AMPK-mediated energy signalling and into the pharmacology of thiopurines and antifolates, whose actions converge on the same node. We offer the prediction as a theoretical target and invite its experimental test.

A. Model equations, parameters and numerics

State and balances. The state is $([\text{AMP}], [\text{ADP}], [\text{ATP}], [\text{For}], [\text{CHO}], [\text{PRPP}], [\text{GAR}], [\text{AICAR}])$, with $[\text{THF}] = F_{\text{tot}} - [\text{CHO}]$ and $c = [\text{CHO}]/(K_c + [\text{CHO}])$. Reversible FTHFS is

$$v_{\text{FTHFS}} = \frac{h}{F_{\text{tot}}(H + [\text{For}])} \left([\text{For}][\text{THF}] - \frac{[\text{CHO}][\text{ADP}][P_i]}{K_{eq}[\text{ATP}]} \right),$$

Table 2

Parameters added to the model of Oizel et al. (2020). Shared parameters retain their published values. Provenance: fitted to Witus et al. (2026) (F), literature (L), or assumed (A).

Symbol	Value	Prov.
K_{dP}	104 μM (PRPP dissociation)	F
K_a	56.4 μM (AMP glue affinity)	F
n	1.33 (Hill coefficient)	F
F_{tot}	0.02 mM (folate pool)	A
K_c	0.002 mM (GART/ATIC K_m , CHO-THF)	A
K_g, K_t	0.01 mM (GART/ATIC K_m , GAR/AICAR)	A
$g_{\text{max}}, t_{\text{max}}$	5 $\times$ demand (GART/ATIC capacity)	A
K_{PRPP}	0.3 mM (PPAT K_m , PRPP)	A
P_{max}	2 $\times$ demand (PPAT capacity)	A
S_{PRPP}	2 $\times$ demand (PRPP supply)	A
k_o	2 h ⁻¹ (PRPP drain)	A
k_{deg}	0.1 h ⁻¹ (GAR/AICAR drain)	A
K_{eq}	40 (FTHFS equilibrium)	L
$[P_i]$	5 mM (orthophosphate)	A

reducing to the published irreversible law as $K_{eq} \rightarrow \infty$. The eight steady-state balances are: formate, $hf_{SCF} + k_F[\text{For}_x] = v_{\text{FTHFS}} + k_F[\text{For}]$; CHO-THF, $v_{\text{FTHFS}} + hf_{S,\text{CHO}} = v_{\text{GART}} + v_{\text{ATIC}} + v_{\text{TYMS}}$; PRPP, $S_{\text{PRPP}} = v_{\text{PPAT}} + (k_o + \mu)[\text{PRPP}]$; GAR, $v_{\text{PPAT}} = v_{\text{GART}} + (\mu + k_{\text{deg}})[\text{GAR}]$; AICAR, $v_{\text{GART}} = v_{\text{ATIC}} + (\mu + k_{\text{deg}})[\text{AICAR}]$; adenine, $\phi v_{\text{ATIC}} = \mu([\text{AMP}] + [\text{ADP}] + [\text{ATP}] + \Pi_{\text{ade}})$; energy, $a \frac{[\text{ATP}]}{A + [\text{ATP}]} = e_g \frac{[\text{ADP}]}{E_g + [\text{ADP}]} + e_o \frac{[\text{ADP}]}{E_o + [\text{ADP}]}$; and adenylate kinase, $[\text{AMP}][\text{ATP}] = K[\text{ADP}]^2$. Here $v_{\text{TYMS}} = \mu[\text{DNA}]/4$ and $\mu = (-m + a[\text{ATP}]/(A + [\text{ATP}]))/\epsilon$.

Numerics. Concentrations are solved in logarithmic coordinates (CHO-THF through a logistic map bounded by F_{tot}), enforcing positivity, by trust-region least squares with restarts from perturbed initial guesses. Each condition is swept forwards and backwards with continuation; agreement to $< 4 \times 10^{-14}$ excludes hysteresis. All reported points converge with residual $< 10^{-13}$.

Data availability

This is a theoretical study and generates no new experimental data. The published data used for calibration and comparison are available in the cited sources: Witus et al. (2026), Strefeler et al. (2025) and Oizel et al. (2020). The model outputs that underlie the figures are provided in the code repository below.

Code availability

The code that implements the model, performs the calibration and produces the figures is available at <https://github.com/av2atgh/Formate-NUDT5>. Every parameter carries an inline provenance tag (fitted, literature, or assumed).

Acknowledgements

Nodes & Links Ltd provided support in the form of salary for Alexei Vazquez, but did not have any additional role in the conceptualization of the study, analysis, decision to publish, or preparation of the manuscript. The code and text were prepared with the assistance of Claude Opus 4.8, an AI assistant developed by Anthropic. A.V. directed the use of the AI assistant, verified all results, and takes full responsibility for the content.

Competing interests

The author declares no competing interests.

References

- Corton, J.M., Gillespie, J.G., Hawley, S.A., Hardie, D.G., 1995. 5-Aminoimidazole-4-carboxamide ribonucleoside: a specific method for activating AMP-activated protein kinase in intact cells? *Eur. J. Biochem.* 229, 558–565.
- Ducker, G.S., Rabinowitz, J.D., 2017. One-carbon metabolism in health and disease. *Cell Metab.* 25, 27–42.
- Hardie, D.G., Ross, F.A., Hawley, S.A., 2012. AMPK: a nutrient and energy sensor that maintains energy homeostasis. *Nat. Rev. Mol. Cell Biol.* 13, 251–262.
- Himes, R.H., Rabinowitz, J.C., 1962. Formyltetrahydrofolate synthetase. III. Studies on the mechanism of the reaction. *J. Biol. Chem.* 237, 2915–2925.
- Meiser, J., Schuster, A., Pietzke, M., et al., 2018. Increased formate overflow is a hallmark of oxidative cancer. *Nat. Commun.* 9, 1368.
- Oizel, K., Tait-Mulder, J., Fernandez-de-Cossio-Diaz, J., et al., 2020. Formate induces a metabolic switch in nucleotide and energy metabolism. *Cell Death Dis.* 11, 310.
- Strefeler, A., Baker, Z.N., Chollet, S., et al., 2025. Uridine-sensitized screening identifies demethoxy-coenzyme Q and NUDT5 as regulators of nucleotide synthesis. *Nat. Metab.* 7, in press. doi:10.1038/s42255-025-01419-2.
- Tibbetts, A.S., Appling, D.R., 2010. Compartmentalization of mammalian folate-mediated one-carbon metabolism. *Annu. Rev. Nutr.* 30, 57–81.
- Witus, S.R., Kober, M.M., Roh, H., et al., 2026. Metabolite glues as a means of purine sensing and chemotherapeutic response. *Nature*, in press. doi:10.1038/s41586-026-10790-3.